Independent validation report: Motor TPL claim frequency GLM

Model under test: legacy Python model pricing_glm: committed reference test predictions (artifacts/reference/pricing_glm, statsmodels
Generated: 2026-11-08T21:43:54.170898+00:00
Status: pending Gate 4

OVERALL RATING: AMBER - Fit for use with conditions: remediate the amber findings.

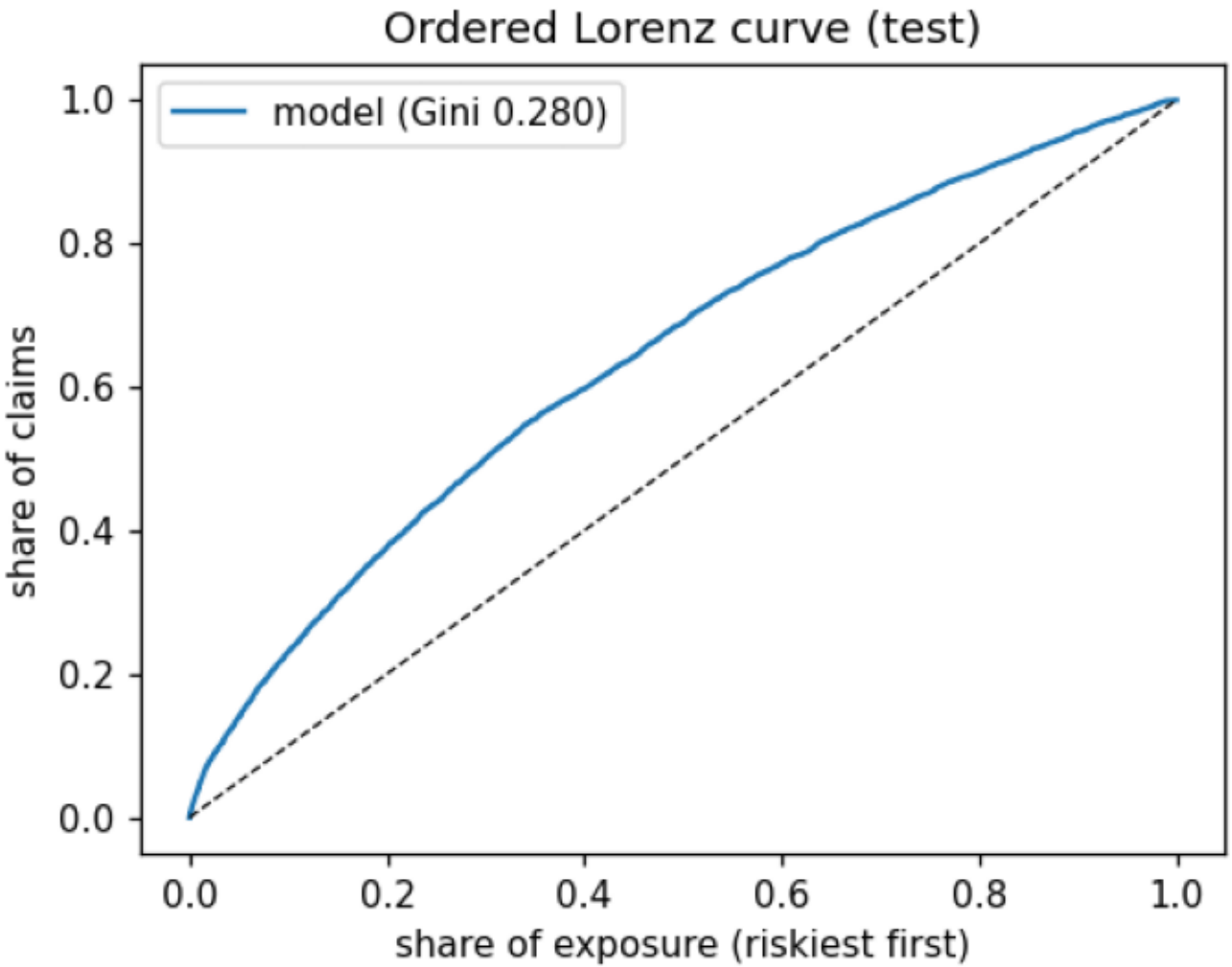
Ratings by area: performance GREEN, calibration AMBER, stability GREEN, fairness AMBER

Executive summary:
The Motor TPL claim frequency GLM is rated amber overall: performance and stability are green, while calibration and fairness are amber. On the held-out test set the model predicts 5,306.9 claims against 5,234 observed, an overall A/E of 0.986, and the test-to-train deviance ratio of 0.986 shows no sign of overfitting. The amber ratings arise from localised miscalibration in the worst predicted-frequency decile (deviation 0.159) and in the worst driver age band (deviation 0.125), both inside the amber tolerance but outside green. There are also documentation gaps: LogDensity is used but undocumented, and Density is documented but not in the fitted model. Remediation should focus on recalibration review and on correcting the documentation before the model is relied on for high-tier use.

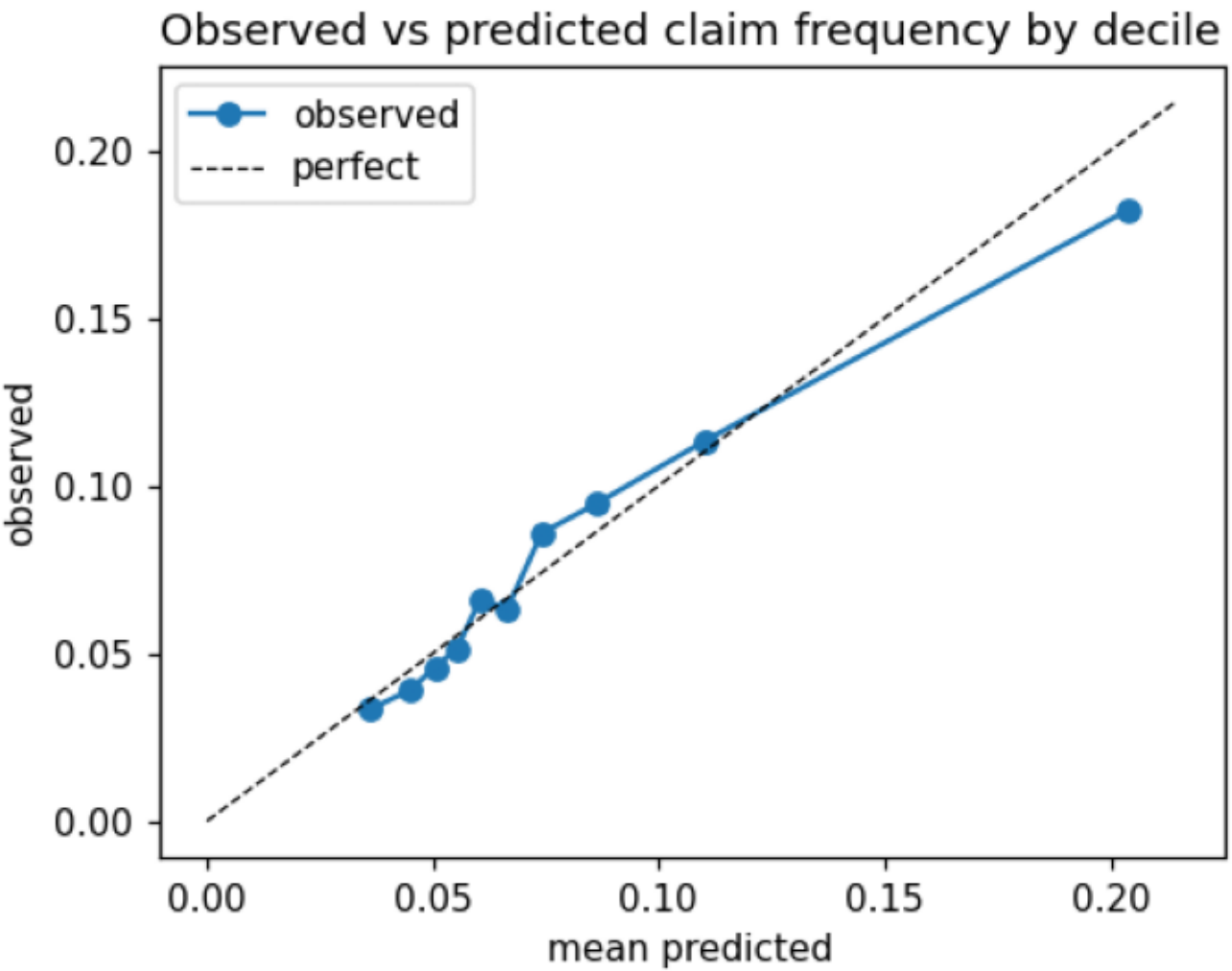
Checks:
[GREEN] performance: Overall actual / expected claims: |A/E - 1| = 0.014 (<= 0.05 green, <= 0.1 amber)
[AMBER] calibration: Worst predicted-frequency decile: |A/E - 1| = 0.159 (<= 0.1 green, <= 0.2 amber)
[GREEN] calibration: Worst Area segment: |A/E - 1| = 0.054 (<= 0.1 green, <= 0.2 amber) segments with exposure >= 1,000
[GREEN] calibration: Worst VehGas segment: |A/E - 1| = 0.019 (<= 0.1 green, <= 0.2 amber) segments with exposure >= 1,000
[GREEN] stability: Input stability: largest feature PSI (train -> test) = 1.14e-04 (<= 0.1 green, <= 0.25 amber) feature Region
[AMBER] fairness: Driver age bands: worst |A/E - 1| = 0.125 (<= 0.1 green, <= 0.2 amber) checks no age band is systematically over- or under-priced

Findings:
[medium] The worst predicted-frequency decile has an A/E deviation of 0.159, outside the green tolerance, although overall A/E is 0.986. Predictions are therefore not uniformly calibrated across the risk range. -> Identify which decile is affected and whether the bias is over- or under-prediction, and test whether recalibration or a revised functional form removes it. Monitor decile A/E in ongoing monitoring with an agreed trigger.
[medium] At least one driver age band shows an A/E deviation of 0.125, suggesting that age band is systematically mispriced relative to experience. -> Review age-band A/E in detail, assess whether the age effect specification (for example banding or non-linearity) needs refinement, and refer the permissibility of using age as a rating factor to the legal and conduct reviewer.
[low] Documentation does not match the fitted model: LogDensity is used but undocumented, while Density is documented as an input but is not in the fitted model. -> Update the model documentation to describe the LogDensity transformation and remove or correct the reference to Density, with owner sign-off.

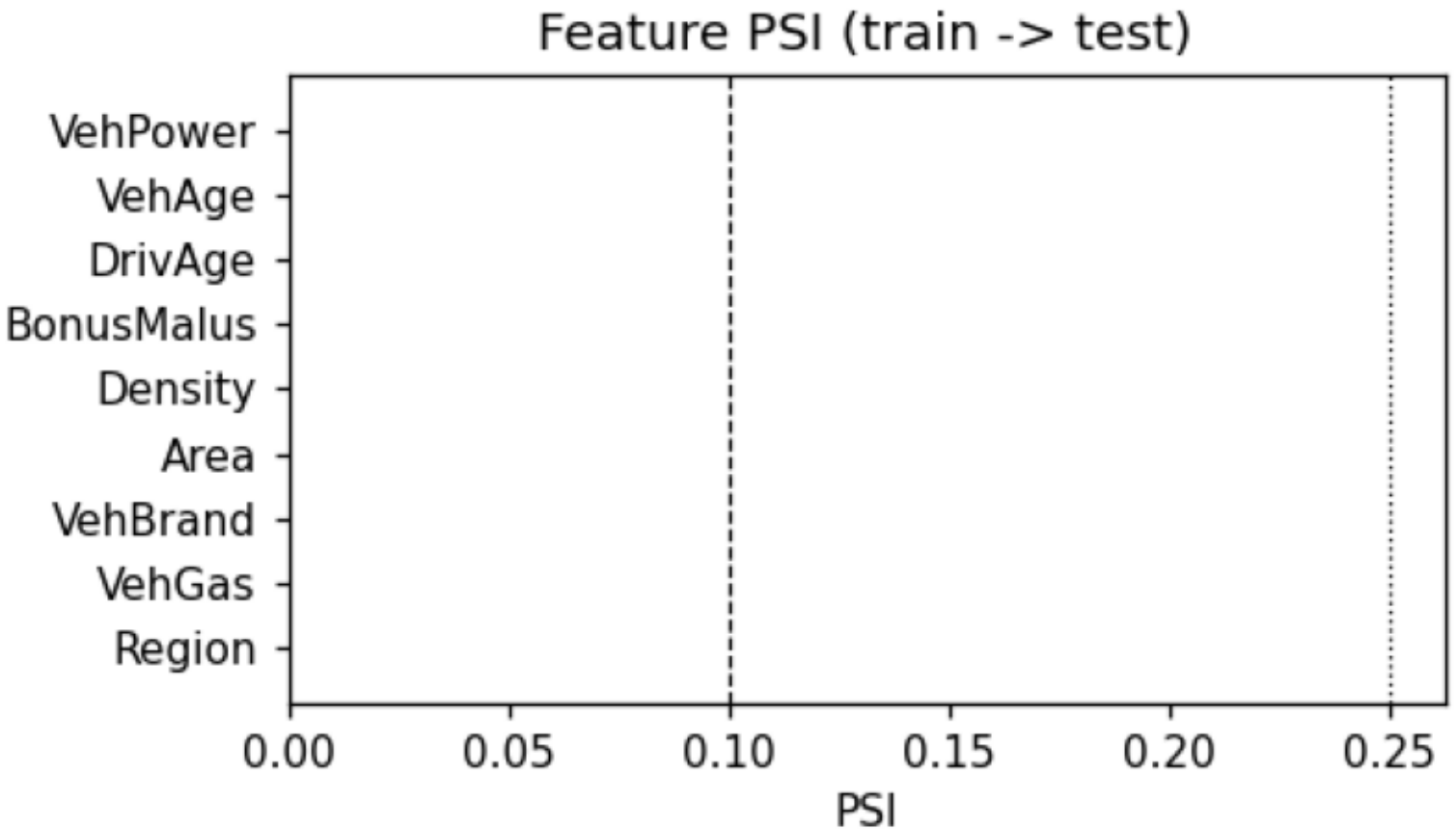
Performance: lorenz curve



Calibration: calibration



Stability: psi by variable



Fairness: ae by driver age

